

Law of Motion of Unemployment

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Law of motion for employment.

$$\dot{l}(t) = f(\theta(t)) [h - l(t)] - \lambda l(t)$$

($\dot{x}(t) \equiv \frac{dx}{dt}$)

change in employment per unit time

workers who lose job per unit time

unemployed workers

job finding rate

workers who find job per unit time

Law of motion of unemployment rate.

Unemployment rate

$$u(t) = 1 - \frac{l(t)}{h}$$

$$\dot{u}(t) = - \frac{\dot{l}(t)}{h} = \lambda \frac{l(t)}{h} - f(\theta(t)) \left[1 - \frac{l(t)}{h} \right]$$

$1 - u(t)$

$u(t)$

$$\dot{u}(t) = \lambda [1 - u(t)] - f(\theta(t)) \cdot u(t)$$

job separations

job findings